

# SKILL - 5

## ARUNTEJ(2520030472)

```
[1]: #Titanic Survival - Full Logistic Regression Pipeline Project
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

#Load the dataset
df = pd.read_csv("titanic.csv")
print(df.head())
print(df.shape)
print(df.columns)

#Target (y) = survived
#to understand the data set
print(df.info())
print(df.describe())
print(df.isnull().sum())#check missing values
#EDA
sns.countplot(x="survived", data=df)
plt.title("Titanic Survival Count")
plt.show()

#Survival by gender
sns.countplot(x="sex", hue="survived", data=df)
plt.title("Survival by Gender")
plt.show()

sns.countplot(x="pclass", hue="survived", data=df)
plt.title("Survival by Passenger Class")
plt.show()

#select the required features
features = ["pclass", "sex", "age", "sibsp", "parch", "fare", "embarked"]
X = df[features].copy()
y = df["survived"]

#for numer is age use median //missing values
X["age"] = X["age"].fillna(X["age"].median())

#for categorical colum use mode
X["embarked"] = X["embarked"].fillna(X["embarked"].mode()[0])
print(X.isnull().sum())
X = pd.get_dummies(X, columns=["sex", "embarked"], drop_first=True)
print(X.head())

#Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
                                                    random_state=42, stratify=y)

scaler = StandardScaler()
#Feature Scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

#Logistic Regression
model = LogisticRegression(max_iter=1000)
model.fit(X_train_scaled, y_train)
print("Model training completed")
y_pred = model.predict(X_test_scaled)
print(y_pred[:20])
```

```

survived  pclass    sex  age  sibsp  parch    fare embarked  class \
0         0         3  male  22.0    1     0   7.2500      S   Third
1         1         1  female 38.0    1     0  71.2833      C   First
2         1         3  female 26.0    0     0   7.9250      S   Third
3         1         1  female 35.0    1     0  53.1000      S   First
4         0         3  male   35.0    0     0   8.0500      S   Third

```

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who  adult_male  deck  embark_town  alive  alone
0   man         True  NaN  Southampton  no    False
1  woman        False  C    Cherbourg   yes    False
2  woman        False  NaN  Southampton  yes    True
3  woman        False  C    Southampton  yes    False
4   man         True  NaN  Southampton  no    True

```

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(891, 15)
```

```

Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
      'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_town',
      'alive', 'alone'],
      dtype='object')

```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 15 columns):
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```

#   Column      Non-Null Count  Dtype
---  ----
0   survived    891 non-null      int64
1   pclass      891 non-null      int64
2   sex         891 non-null      object
3   age         714 non-null      float64
4   sibsp       891 non-null      int64
5   parch       891 non-null      int64
6   fare        891 non-null      float64
7   embarked    889 non-null      object
8   class       891 non-null      object
9   who         891 non-null      object
10  adult_male   891 non-null      bool
11  deck         203 non-null      object
12  embark_town  889 non-null      object
13  alive        891 non-null      object
14  alone        891 non-null      bool

```

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dtypes: bool(2), float64(2), int64(4), object(7)
```

```
memory usage: 92.4+ KB
```

```
None
```

```

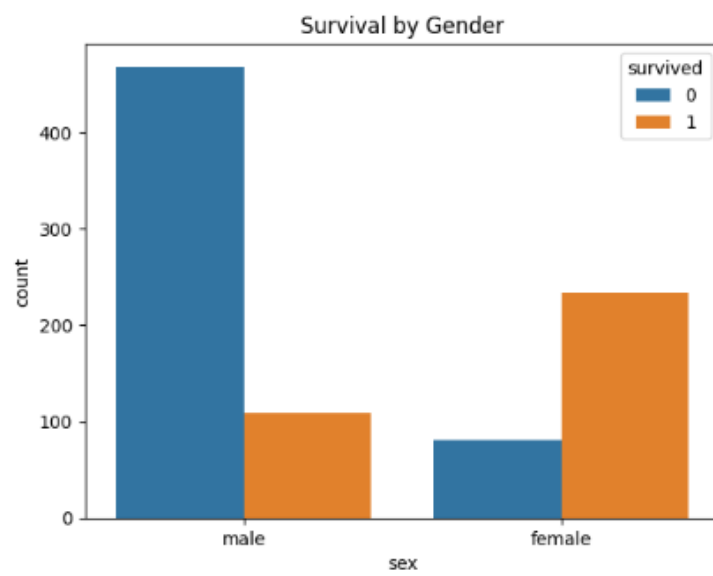
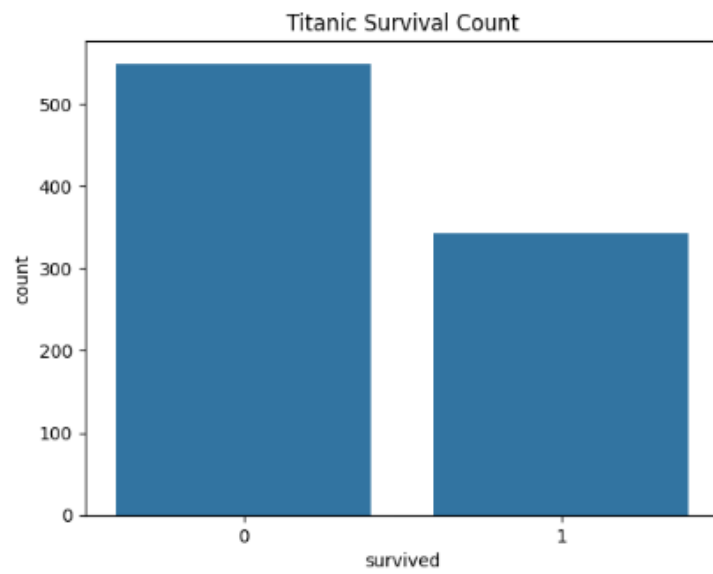
survived  pclass    age  sibsp    parch    fare
count  891.000000  891.000000  714.000000  891.000000  891.000000  891.000000
mean    0.383838    2.308642   29.699118    0.523008    0.381594   32.204208
std     0.486592    0.836071   14.526497    1.102743    0.806057   49.693429
min     0.000000    1.000000    0.420000    0.000000    0.000000    0.000000
25%     0.000000    2.000000   20.125000    0.000000    0.000000    7.910400
50%     0.000000    3.000000   28.000000    0.000000    0.000000   14.454200
75%     1.000000    3.000000   38.000000    1.000000    0.000000   31.000000
max     1.000000    3.000000   80.000000    8.000000    6.000000  512.329200

```

```

survived    0
pclass      0
sex          0
age         177
sibsp       0
parch       0
fare        0
embarked    2
class       0
who         0
adult_male   0
deck        688
embark_town  2
alive       0
alone       0
dtype: int64

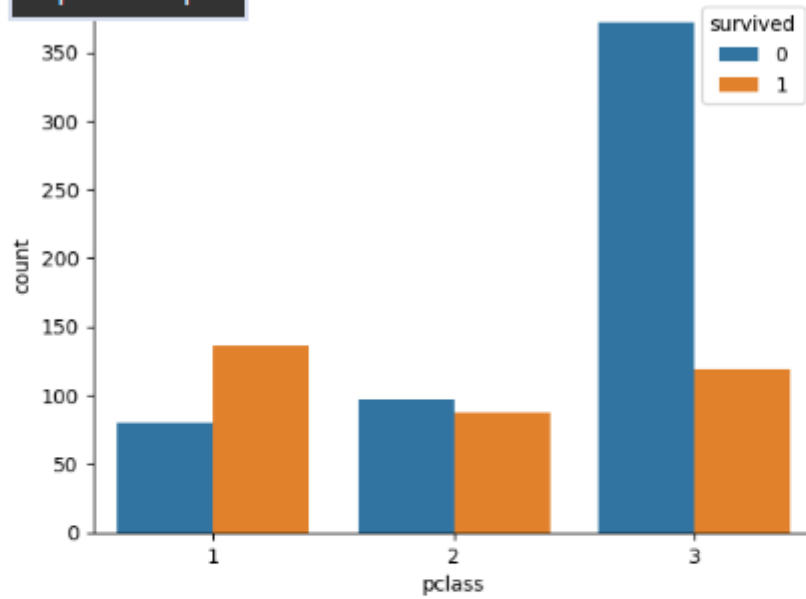
```



sex

Expand Output

Survival by Passenger Class



```
pclass    0
sex       0
age       0
sibsp     0
parch     0
fare      0
embarked  0
dtype: int64
  pclass  age  sibsp  parch   fare  sex_male  embarked_Q  embarked_S
0      3  22.0     1     0   7.2500      True      False      True
1      1  38.0     1     0  71.2833     False      False     False
2      3  26.0     0     0   7.9250     False      False     True
3      1  35.0     1     0  53.1000     False      False     True
4      3  35.0     0     0   8.0500      True      False     True
Model training completed
[0 0 0 0 1 0 1 0 0 0 0 0 0 1 0 0 0 0 0 0 1]
```